

Lecture 8

Tabular (Quine–McCluskey) Minimization

Implicants in dash notation · combining to prime implicants · the prime implicant chart · essential primes · cyclic charts and Petrick's method

1	0	1	0
1	0	1	1
1	0	1	-

$$m_{10} + m_{11} = AB'C$$

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Course supplement: tabular minimization

Today's roadmap

1

Why tabular?

An algorithm, not a picture

2

Prime implicants

Combine, then combine again

3

The PI chart

Essentials first

4

Cyclic charts

Petrick's method

By the end of class you can...

- Write implicants in dash notation
- Find all prime implicants by combining tables
- Build a prime implicant chart and pick the essentials
- Finish a cover when essentials are not enough
- Solve a cyclic chart with Petrick's method

Agenda for today's session

Intro

Why

Step 1

Break

Step 2

Practice

Cyclic

Wrap

0:00

0:05

0:15

0:40

0:50

1:10

1:25

1:40

1:50

Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:15	Why a tabular method; dash notation	—
0:15 – 0:40	Finding the prime implicants	Example 1
0:40 – 0:50	Break	
0:50 – 1:10	Prime implicant chart; essentials	Example 1 cover
1:10 – 1:25	Full problem in pairs	Your turn
1:25 – 1:40	Cyclic charts and Petrick's method	Example 2
1:40 – 1:50	Concept check and wrap-up	Exam 1 preview

Why a tabular method?

Limits of K-maps

Adjacency is hard to see past 4 variables and impractical past 6.

Choosing groups by eye can miss a better cover.

The tabular method

Same theorem as the map, $xy + xy' = x$, applied systematically.

Works for any number of variables and is easy to program.

- 1 **Step 1:** list minterms and don't-cares in binary, grouped by the number of 1s.
- 2 Combine terms in neighbouring groups that differ in one bit; repeat until nothing combines.
- 3 Terms never combined are the **prime implicants**.
- 4 **Step 2:** build the chart; take the **essential** primes.
- 5 Cover the remaining minterms with the fewest extra primes.

Implicants in dash notation

Terms	Patterns	Result	Term
m_{10}, m_{11}	1010, 1011	101–	$AB'C$
m_8, m_{12}	1000, 1100	1–00	$AC'D'$
101–, 111–	differ in bit B	1–1–	AC
m_4, m_9	0100, 1001	—	3 bits differ
10–1, 1–10	dashes differ	—	cannot combine

The combining rule

Two terms combine when they differ in exactly one bit and their dashes are in the same places.

The differing bit becomes a dash: one fewer literal.

Only neighbouring groups (one more 1) can differ in one bit, so only those are compared.

Example 1 — List by number of 1s

EXAMPLE

Minimize $F(A, B, C, D) = \Sigma m(4, 8, 10, 11, 12, 15) + d(9, 14)$ by the tabular method.

1s	Minterms	Pattern	
1	4	0100	✓
	8	1000	✓
2	(9)	1001	✓
	10	1010	✓
	12	1100	✓
3	11	1011	✓
	(14)	1110	✓
4	15	1111	✓

- 1 Include the don't-cares 9 and 14: they may help combining.
- 2 Write each term in binary, A as the MSB.
- 3 Group by the number of 1s: 1, 2, 3 and 4.
- 4 Compare every term with every term in the next group.
- 5 Mark ✓ each term that combines; the rest are primes.

Example 1 — Size-2 implicants

1s	Minterms	Pattern	
1	4,12	-100	PI
	8,9	100-	✓
	8,10	10-0	✓
	8,12	1-00	✓
2	9,11	10-1	✓
	10,11	101-	✓
	10,14	1-10	✓
	12,14	11-0	✓
3	11,15	1-11	✓
	14,15	111-	✓

- Group 1–2: $4,12 \rightarrow -100$; $8,9 \rightarrow 100-$; $8,10 \rightarrow 10-0$; $8,12 \rightarrow 1-00$
- Group 2–3: $9,11$; $10,11$; $10,14$; $12,14$
- Group 3–4: $11,15 \rightarrow 1-11$; $14,15 \rightarrow 111-$
- -100 combines with nothing: prime $BC'D'$
- Every other size-2 term combines in the next column

Example 1 — Size-4 and the primes

1s	Minterms	Pattern	
1	8,9,10,11	10--	PI
	8,10,12,14	1--0	PI
2	10,11,14,15	1-1-	PI

Prime implicant	Pattern	Covers
$BC'D'$	-100	4, 12
AB'	10--	8, 9, 10, 11
AD'	1--0	8, 10, 12, 14
AC	1-1-	10, 11, 14, 15

- 1 Combine size-2 terms whose dashes line up
- 2 10-0 with 10-1 gives **10--**, and so on
- 3 The same size-4 term appears twice: list it once
- 4 No size-4 terms combine further
- 5 Four prime implicants in all

Break

10 minutes — we resume at the 0:50 mark

Next: the prime implicant chart

The prime implicant chart

Prime	Term	4	8	10	11	12	15
-100	$BC'D'$	x				x	
10--	AB'		x	x	x		
1--0	AD'		x	x		x	
1-1-	AC			x	x		x

Reading the chart

Columns: the minterms only (no don't-cares).

A column with a single x makes its row essential.

- 1 Column 4 has one x: $BC'D'$ is essential
- 2 Column 15 has one x: AC is essential
- 3 They cover 4, 12 and 10, 11, 15
- 4 Only minterm 8 is left
- 5 AB' or AD' covers it: either gives a minimum

Example 1 — The minimum cover

Check on the map

AB \ CD		CD			
		00	01	11	10
AB	00	m0	m1	m3	m2
	01	m4	m5	m7	m6
	11	m12	m13	m15	m14
	10	m8	m9	m11	m10

00					
01	1				
11	1		1	X	
10	1	X	1	1	

1 Essentials: $BC'D'$ and AC

2 Add AB' (or AD') for minterm 8

3
$$F = BC'D' + AC + AB'$$

4 or $F = BC'D' + AC + AD'$: same cost

5 7 literals, 3 terms: matches the map

Your turn — A full problem

YOUR TURN

≈ 15 min

Minimize $F(A, B, C, D) = \Sigma m(0, 1, 2, 5, 6, 7, 8, 9, 10, 14)$ by the tabular method.

- (a) Build the combining columns and list the prime implicants.
- (b) Draw the prime implicant chart and mark the essentials.
- (c) Write the minimum sum of products.

Six prime implicants exist. Check your answer on a K-map.

Solution — Combining columns

SOLUTION

1s	Minterms	Pattern	
0	0	0000	✓
1	1	0001	✓
	2	0010	✓
	8	1000	✓
2	5	0101	✓
	6	0110	✓
	9	1001	✓
	10	1010	✓
3	7	0111	✓
	14	1110	✓

1s	Minterms	Pattern	
0	0,1	000-	✓
	0,2	00-0	✓
	0,8	-000	✓
1	1,5	0-01	PI
	1,9	-001	✓
	2,6	0-10	✓
	2,10	-010	✓
	8,9	100-	✓
	8,10	10-0	✓
2	5,7	01-1	PI
	6,7	011-	PI
	6,14	-110	✓
	10,14	1-10	✓

1s	Minterms	Pattern	
0	0,1,8,9	-00-	PI
	0,2,8,10	-0-0	PI
1	2,6,10,14	--10	PI

- Column 1: the ten minterms, grouped by their number of 1s.
- Column 2: join terms in neighbouring groups that differ in one bit; ✓ both.
- Column 3: join column-2 terms whose dashes line up.
- Rows marked PI never combined: they are the prime implicants.

Solution — The prime implicants

SOLUTION

Prime implicant	Pattern	Covers	Column
$A'C'D$	0-01	1, 5	size 2
$A'BD$	01-1	5, 7	size 2
$A'BC$	011-	6, 7	size 2
$B'C'$	-00-	0, 1, 8, 9	size 4
$B'D'$	-0-0	0, 2, 8, 10	size 4
CD'	--10	2, 6, 10, 14	size 4

- 1 Column sizes: 10, 13, 3
- 2 Three size-4 primes: $B'C'$, $B'D'$, CD'
- 3 Three size-2 primes remain unused
- 4 $A'C'D$, $A'BD$, $A'BC$
- 5 Six primes in all

Solution — Chart and cover

SOLUTION

Prime	Term	0	1	2	5	6	7	8	9	10	14
0-01	$A'C'D$		x		x						
01-1	$A'BD$				x		x				
011-	$A'BC$					x	x				
-00-	$B'C'$	x	x					x	x		
-0-0	$B'D'$	x		x				x		x	
--10	CD'			x		x				x	x

1 Columns 9 and 14 each have a single x.

2 So $B'C'$ and CD' are **essential**.

3 They cover 0, 1, 2, 6, 8, 9, 10, 14.

4 Left to cover: minterms 5 and 7.

5 $A'BD$ covers both ($A'C'D$ and $A'BC$ cover one each).

6 $F = B'C' + CD' + A'BD$ (7 literals)

Cyclic charts and Petrick's method

Cyclic chart

Every column has at least two x: there are no essential primes to start from.

Petrick's method

Name the primes $P_1, P_2, \dots$. For each minterm write the OR of the primes that cover it.

AND all those sums, multiply out, and absorb: each product is a complete cover.

Pick the product with the fewest (then cheapest) primes.

Useful reductions

$$(P_1 + P_2)(P_1 + P_3) = P_1 + P_2P_3$$

Row dominance: drop a prime whose x are a subset of another's at no lower cost.

Column dominance: drop a minterm column whose x include another column's.

CAD tools use these steps with branching for large problems.

Example 2 — A cyclic chart

EXAMPLE

Minimize $F(A, B, C) = \Sigma m(0, 1, 2, 5, 6, 7)$.

	Prime	Term	0	1	2	5	6	7
P1	00-	$A'B'$	x	x				
P2	0-0	$A'C'$	x		x			
P3	-01	$B'C$		x		x		
P4	-10	BC'			x		x	
P5	1-1	AC				x		x
P6	11-	AB					x	x

- 1 Six primes, each covering 2 minterms
- 2 Every column has two x: no essentials
- 3 Each column needs one of its two primes:
- 4
$$P = (P_1 + P_2)(P_1 + P_3)(P_2 + P_4)(P_3 + P_5)(P_4 + P_6)(P_5 + P_6)$$
- 5 Multiply out and absorb (next slide)

Example 2 — Petrick's method

- 1 One sum per minterm: 0 ($P_1 + P_2$), 1 ($P_1 + P_3$), 2 ($P_2 + P_4$), 5 ($P_3 + P_5$), 6 ($P_4 + P_6$), 7 ($P_5 + P_6$)
- 2 Pair sums that share a prime: $(P_1 + P_2)(P_1 + P_3) = P_1 + P_2P_3$
- 3 $(P_2 + P_4)(P_4 + P_6) = P_4 + P_2P_6$; $(P_3 + P_5)(P_5 + P_6) = P_5 + P_3P_6$
- 4 Multiply the first two: $P_1P_4 + P_1P_2P_6 + P_2P_3P_4 + P_2P_3P_6$
- 5 Multiply by $(P_5 + P_3P_6)$ and use $XX = X$:
 $P_1P_4P_5 + P_1P_3P_4P_6 + P_1P_2P_5P_6 + P_1P_2P_3P_6 + P_2P_3P_4P_5 + P_2P_3P_4P_6 + P_2P_3P_5P_6 + P_2P_3P_6$
- 6 Absorb: $P_2P_3P_6$ absorbs every product that contains it
- 7 Left: $P_1P_4P_5 + P_2P_3P_6 + P_1P_3P_4P_6 + P_1P_2P_5P_6 + P_2P_3P_4P_5$
- 8 Fewest primes: $P_1P_4P_5$ and $P_2P_3P_6$, 3 each

Example 2 — The two covers

Two minimum covers

A \ BC		BC			
		00	01	11	10
A	0	1	1		1
	1		1	1	1

$$A'B' + BC' + AC$$

A \ BC		BC			
		00	01	11	10
A	0	1	1		1
	1		1	1	1

$$A'C' + B'C + AB$$

1 $P_1P_4P_5: P_1 = A'B', P_4 = BC', P_5 = AC$

2 $F = A'B' + BC' + AC$

3 $P_2P_3P_6: P_2 = A'C', P_3 = B'C, P_6 = AB$

4 $F = A'C' + B'C + AB$

5 Each map: every 1 is covered by the three groups ✓

6 Both: 3 terms, 6 literals: either is a minimum

Tabular method or K-map?

	K-map	Tabular method
Variables	up to 4 (5–6 with effort)	any number
Method	visual pattern finding	systematic algorithm
Don't-cares	used when grouping	used in step 1 only
Prime implicants	found by eye	all found, guaranteed
Choosing a cover	by eye	chart, then Petrick
Best for	hand work, insight	computers, large problems

The number of prime implicants can grow exponentially, so real CAD tools add heuristics.

Concept check

1

Why are only neighbouring groups (by count of 1s) compared?

2

Why are don't-cares used in step 1 but not in the chart?

3

What makes a prime implicant essential?

4

What does it mean when every chart column has two or more $\times$?

Key takeaways

Combine

$$xy + xy' = x$$

Dash notation

$$101- = AB'C$$

Prime implicant

never combined

Essential

only $\times$ in a column

Cyclic chart

Petrick's method

Don't-cares

step 1 only

Same theorem as the K-map, applied by algorithm.

Homeworks #4 and #5 due at the start of Lecture 9

This afternoon's lab: Lab C — Gate-level Verilog, propagation delay, testbenches

Next: review for Exam 1